



## B4 Gas exchange

B · Form and function · IB Diploma · Concept guide and original exam-style practice

### Understand the concept

Efficient gas exchange surfaces, such as alveoli in mammalian lungs, gills in fish, and leaves in plants, share common features: a large surface area, a short diffusion distance (thin, often single-cell-thick walls), a moist surface for gases to dissolve in, and a steep concentration gradient maintained by ventilation and/or blood flow. In fish gills, blood and water flow in opposite directions (countercurrent exchange), maintaining a diffusion gradient for oxygen along the entire length of the gill lamellae; in mammals, contraction of the diaphragm and intercostal muscles changes thoracic volume and pressure, drawing air into and out of the lungs (ventilation).

### Worked example: follow the reasoning

Comparing countercurrent and concurrent gas exchange in fish gills. In concurrent flow, blood and water move in the same direction; oxygen diffuses from water to blood until their oxygen concentrations equalise partway along the gill, after which no further net diffusion occurs, so maximum extraction is around 50%. In countercurrent flow, water and blood move in opposite directions; blood with the lowest oxygen concentration meets water that has already lost some oxygen but still has more than the blood, and this relationship holds all along the gill lamella. Conclusion: because a favourable concentration gradient for oxygen is maintained along the whole exchange surface, countercurrent flow allows fish to extract up to 80-90% of the dissolved oxygen from water, far more than concurrent flow would allow.

### Quick check · multiple choice

Which combination of features makes the alveoli of mammalian lungs an efficient gas exchange surface?

- A Thick walls and a poor blood supply
- B A large total surface area, thin walls, and a dense capillary network
- C A small surface area but very thick walls
- D Low ventilation and low blood flow

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 During vigorous exercise, a person's breathing rate and depth increase substantially. [8]
  - (a) Describe the changes in thoracic volume and pressure that occur during inhalation. [3]
  - (b) Explain, in terms of chemoreceptors and CO<sub>2</sub> concentration, why breathing rate increases during exercise. [3]
  - (c) Explain why an increased breathing rate helps maintain a steep diffusion gradient for oxygen at the alveoli. [2]

Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [8]

(a) The diaphragm and external intercostal muscles contract; the diaphragm flattens and the rib cage moves up and out, increasing the volume of the thoracic cavity; this decreases the air pressure inside the lungs below atmospheric pressure, so air flows into the lungs down the pressure gradient.

(b) Increased respiration during exercise produces more CO<sub>2</sub>, which lowers blood pH; chemoreceptors (mainly in the medulla oblongata, also in the aortic/carotid bodies) detect this change and send nerve impulses that increase the rate and depth of breathing (ventilation rate) to expel CO<sub>2</sub> faster and restore homeostasis.

(c) Faster, deeper breathing brings fresh air with a high oxygen concentration into the alveoli more frequently, and removes CO<sub>2</sub>-rich air faster, maintaining a large difference in oxygen concentration between alveolar air and the blood in the capillaries, which maximises the rate of diffusion of oxygen into the blood.

### Quick check answer

B. Millions of alveoli give a huge total surface area, their walls are one cell thick (short diffusion distance), and a dense capillary network maintains a steep concentration gradient for diffusion.

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.